Retail AI Suite — Store Manager & Staff User Manual
(Customer Footfall Analytics & Touchless Staff Attendance)

📖 Welcome & Overview

Welcome to the **OmniRetail AI Suite**. This desktop software is designed for **Store Managers, Supervisors, and HR Personnel**. It operates automatically using your store's security and attendance cameras to provide:

1. **Live Customer Traffic (Footfall)**: Real-time count of customers entering, exiting, and currently shopping inside the store.
2. **Customer Demographics**: Automatic detection of customer age groups and gender to understand who visits your store.
3. **Bag Exit Monitoring**: Alerts and counts when customers exit with shopping bags.
4. **Touchless Staff Attendance (VisionGuard AI)**: Automatic face-recognition check-in and check-out for store employees without touching any fingerprint scanner.
5. **Customer-to-Staff Ratio**: Shows how many shoppers each staff member is currently serving.

👥 Who Uses This Software?

• Store Managers & Shift In-Charges: Monitor store occupancy, customer rush hours, and staff availability.
• HR & Payroll Admins: Enroll new employees, review daily attendance logs, and export work hours.
• Loss Prevention / Security Staff: Verify customer movement and bag exits at checkout.

🚀 Part 1: Starting the Software

1. Look for the application shortcut on the Desktop: **OmniRetail AI Suite** (or run `run_unified_app.bat`).
2. Double-click to launch.
3. The main dashboard will open in full-screen dark theme.

> [!TIP]
> **No Daily Setup Needed**: The software automatically connects to your store cameras, loads today's records, and begins monitoring immediately upon launch.

🖥️ Part 2: Main Screen Navigation

At the top of the window, you will find four main navigation buttons:

Tab	What It Does
📊 **Store Executive Dashboard**	The primary daily overview combining customer traffic and staff on duty.
👥 **Customer Analytics & Demographics**	Detailed live camera feed showing customer tracking lines, age, and gender.
👤 **Staff Attendance & VisionGuard**	Staff check-in screen, daily attendance logs, and new employee enrollment.
⚙️ Central Settings & Diagnostics	Store ID, camera connection checks, and system diagnostics.

In the **top-right corner**, you will see status lights:
• 🟢 **Retail: Active** — Customer tracking camera is operating normally.
• 🟢 **VisionGuard: Active** — Staff attendance camera is ready and scanning.
• 🔴 **Reconnecting...** — Camera network cable is unplugged or camera is offline.

📊 Part 3: Understanding the Executive Dashboard (Tab 1)
This is the screen you will keep open throughout the business day. It displays **6 live KPI cards**:
This is the screen you will keep open throughout the business day. It displays **6 live KPI cards**:

1. Customer Inflow (Green Card)
• **What it means**: Total verified customers who entered the store today.
• **Smart Filter**: If someone steps inside for 2 seconds and turns back immediately, the system does not count them as a customer.

2. Customer Outflow (Blue Card)
• **What it means**: Total customers who finished shopping and left the store today.

3. Current Occupancy (Purple Card)
• **What it means**: How many customers are **inside your store right now**.
* Calculated automatically as: `Customer Inflow - Customer Outflow`.

4. Staff On Duty (Amber Card)
• **What it means**: Number of enrolled staff members who have clocked in today (e.g., `8 / 10` means 8 out of 10 enrolled employees are on duty).

5. Patron : Staff Ratio (Cyan Card)
• **What it means**: Service load factor.
• **Example**: If there are 30 shoppers in the store and 6 staff on duty, the ratio displays `5.0 : 1` (1 employee per 5 shoppers).
• **Why it matters**: If this number jumps above `15.0 : 1`, the store is understaffed for the current customer rush.

6. Bag Exit Crossings (Amber Card)
• **What it means**: Count of customers confirmed leaving through the checkout lane carrying shopping bags.

👥 Part 4: Customer Analytics & Demographics (Tab 2)

Click on 👥 **Customer Analytics & Demographics** to see the detailed camera view:

1. **Live Camera Feed**:
   * Each person walking in is highlighted with a tracking box.
   * A label appears showing their estimated demographic profile:
     e.g., `Male, ~28y (Young Adult)` or `Female, ~34y (Adult)`.
2. **Virtual Gate Lines**:
   • **Yellow Line**: Customer entry / exit counting line.
   • **Green Line**: Demographics reading boundary.
   • **Cyan Line**: Shopping bag exit boundary.
3. **Controls**:
   • **Start / Stop Monitoring**: Pauses or resumes customer tracking.
   • **Calibrate Gates**: Used during initial setup if the store entrance changes.

👤 Part 5: Staff Attendance & Employee Management (Tab 3)

Click on 👤 **Staff Attendance & VisionGuard** to manage employee attendance:

How Staff Clock In / Out (Touchless Face Recognition):
1. The employee steps in front of the attendance camera (at a distance of 0.5 to 1.5 meters).
2. They look at the camera for **1 to 2 seconds**.
3. The screen will turn **Green** with an on-screen confirmation badge:
   • **"Welcome, [Employee Name]! Clock-In Recorded at 09:02 AM"**
4. An audible confirmation chime sounds, and the table updates instantly.

> [!NOTE]

> **Anti-Duplicate Rule**: If an employee walks past the camera multiple times within 5 minutes, the system will not create duplicate entries.

How to Enroll a New Employee:

When a new team member joins your store:

1. Click on the **"Enroll Staff"** button inside Tab 3.
2. Fill in the details:
   • **Employee ID**: (e.g., `EMP-1042`)
   • **Full Name**: (e.g., `Rahim Ahmed`)
   • **Department / Role**: (e.g., `Sales Associate`, `Cashier`)
3. Choose one of two photo options:
   • **Option A (Live Camera Snapshot)**: Have the employee stand centered in front of the camera and click **"Capture Face"**.
   • **Option B (Upload Photo)**: Click **"Browse Image"** and select a clear passport-style photo from your computer.
4. Click "Save & Register Employee".
5. The employee can now clock in immediately.

How to View & Export Daily Attendance Reports:

1. In Tab 3, review the table listing all employee check-ins for the day:
   * Columns include: `Employee ID`, `Name`, `Check-In Time`, `Check-Out Time`, `Status`, `Confidence`.
2. To save the report:
   • Click **"Export to Excel / CSV"**.
   * Choose a folder on your computer (e.g., `Desktop` or `Documents`).
   * The file is ready to email to HR or head office.

⚙️ Part 6: Daily Best Practices & Maintenance

Time	Action
Morning (Store Opening)	Turn on the computer monitor. Check that both status indicators at the top show 🟢 **Green**.
During Day	Keep the **Store Executive Dashboard (Tab 1)** open. Monitor customer occupancy and staff ratios.

| **Evening (Store Closing)** | Click **Export Report** on Tab 3 if your regional office requires daily attendance records. |
| **Shutdown** | Click the red **"⏹ Exit"** button in the top-right corner to close the application cleanly. |

❓ Part 7: Troubleshooting (FAQ)

Q1: The top status bar says "🔴 Reconnecting..."
• **Cause**: The computer lost connection to the camera.
• **Fix**:
  1. Check if the camera's network cable is plugged in firmly.
  2. Verify that the store Wi-Fi/switch router is powered on.
  3. The system will automatically reconnect as soon as the network cable is reattached (no need to restart the app).

Q2: A staff member's face is not being recognized
• **Cause**: Poor lighting, extreme face angle, or cap/mask blocking the face.
• **Fix**:
  1. Ask the employee to remove sunglasses, hats, or masks covering the eyes/nose.
  2. Ensure the employee is standing directly facing the camera in well-lit conditions.
  3. If recognition still fails, go to Tab 3 and update their profile with a clearer photo.

Q3: Customer numbers seem too low or are not counting
• **Cause**: The camera lens is dirty, tilted, or someone blocked the camera view.
• **Fix**:
  1. Clean the camera lens with a microfiber cloth.
  2. Ensure the entrance doorway is not obstructed by tall cardboard banners or promotional posters that block the camera's line of sight.

Q4: Can someone clock in using a photo from their mobile phone?
• **No**. The system includes a built-in **Liveness & Quality Filter** that rejects flat screens, photos, and blurry images.

📞 Support & Helpdesk

For technical assistance, store ID changes, or camera adjustments, contact:
• **Retail IT Support Desk**: `ai-support@prangroup.com`
• **Central API Gateway**: `http://ai.prangroup.com:8021`